Fourier Qualia Wavescapes: Hierarchical Analyses of Set Class Quality and Ambiguity

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and Fabian C. Moss











...Fourier Qualia Wavescapes: a triangular and hierarchical representation of a piece's harmonic qualities...

...qualia (or quale): set class quality such as octatonic...

...overview...

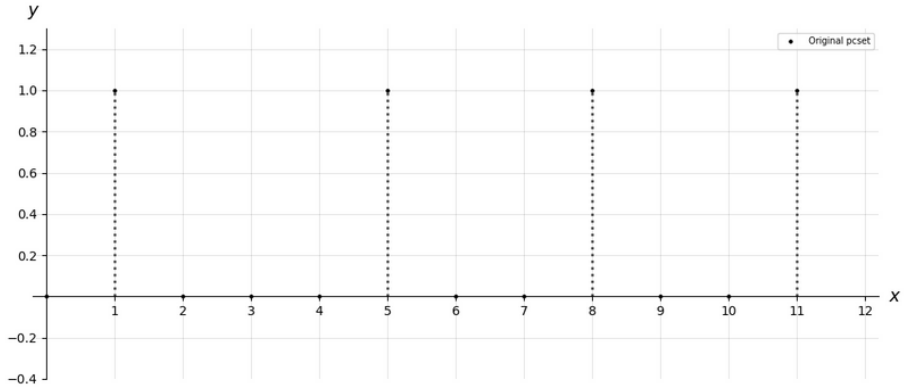
...method to map qualia ambiguity to wavescapes...

1. The discrete Fourier transform of pitch-class sets.

- Signal X is dominant 7th chord with the root in Db.

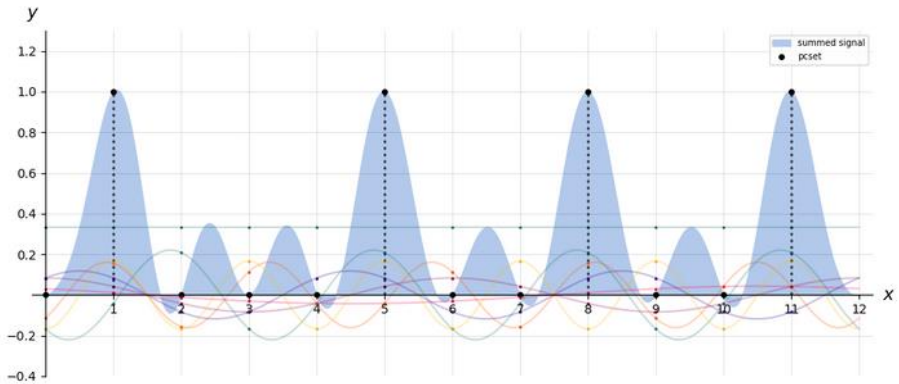
- X = [0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1]

- The DFT of a pcset [1, 5] produces 6 non-trivial coefficients [7, 8, 11].



- *Discrete representation of X*

$\hat{f}_0 + \hat{f}_1 + \cdots + \hat{f}_{11}$

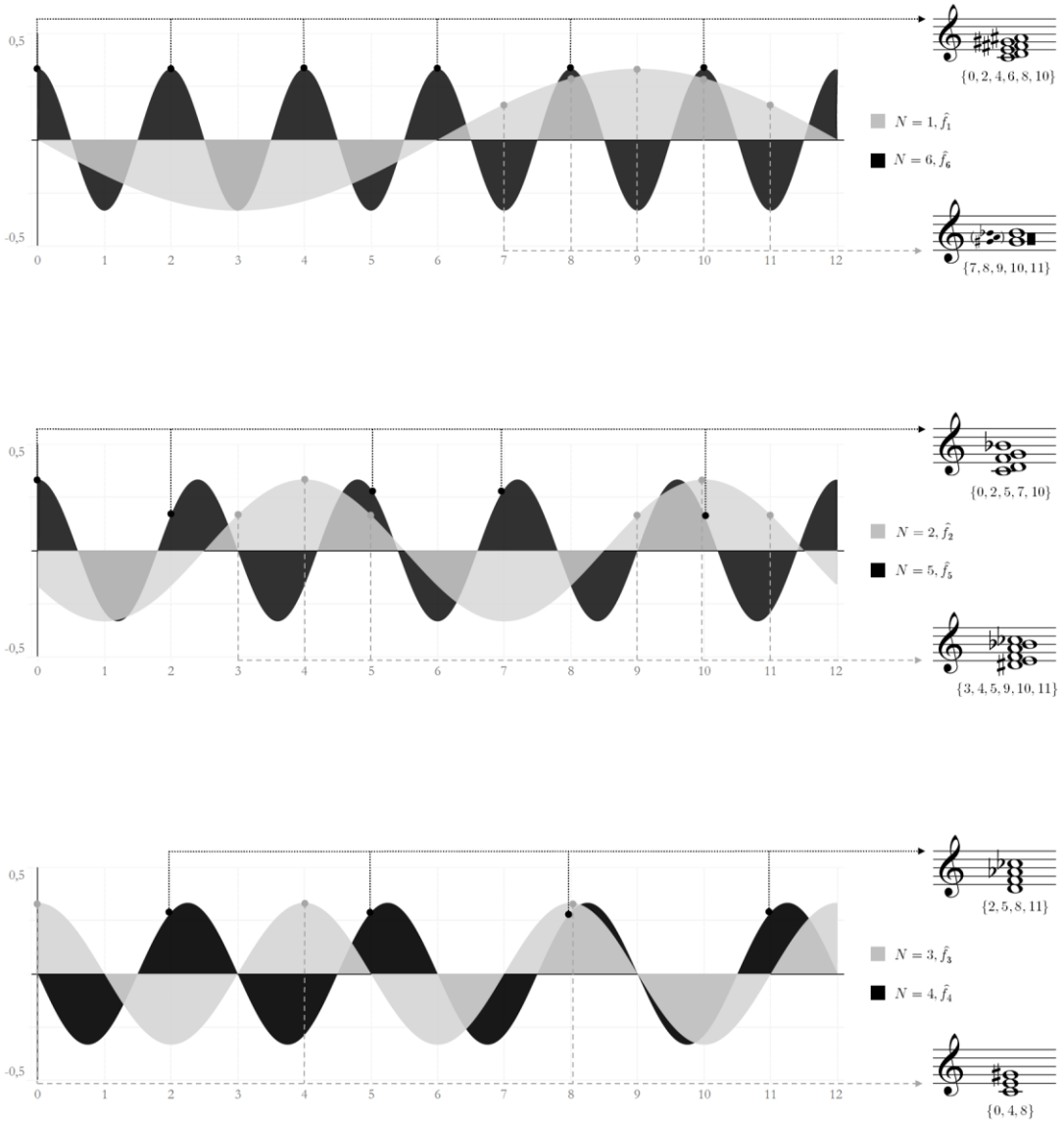


- *DFT coefficients for chord X.*

1. The discrete Fourier transform of pitch-class sets.

- Association between DFT coefficients and harmonic qualities [1, 2, 8, 11]:

$|\hat{f}_1|$ - clustered quality;
$|\hat{f}_2|$ - diadicity;
$|\hat{f}_3|$ - triadicity;
$|\hat{f}_4|$ - octatonicity;
$|\hat{f}_5|$ - diatonicity;
$|\hat{f}_6|$ - whole-tone quality.

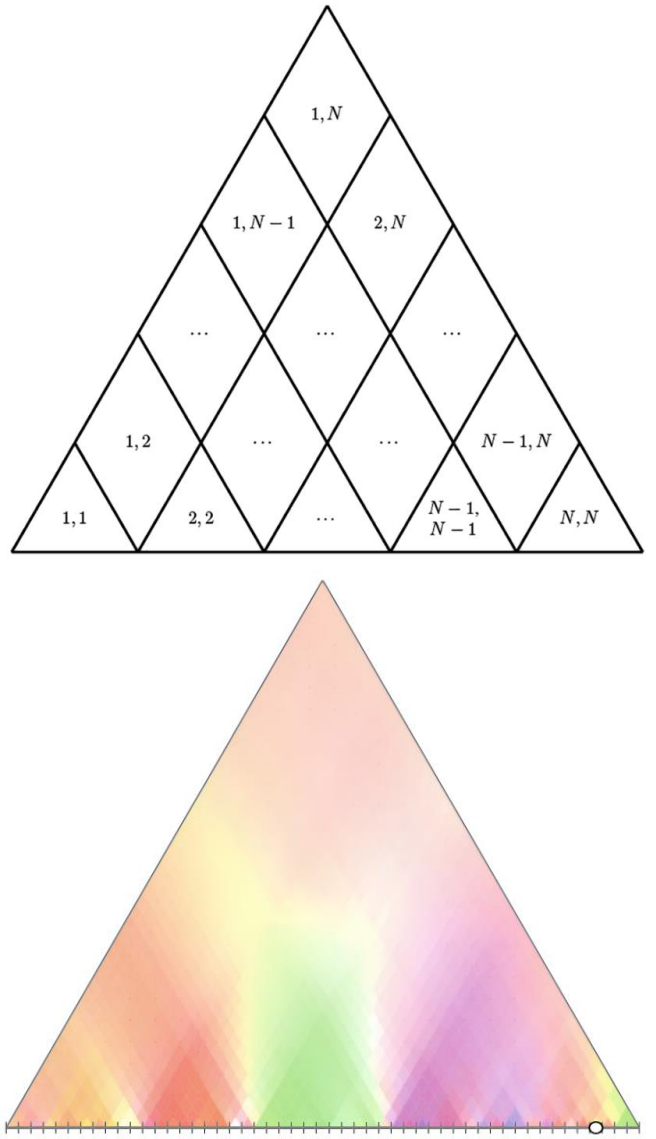


- *DFT coefficients' sinusoids and prototypical chords [6].*

2. Keyscapes + DFT = Wavescapes.

- Wavescapes [10] constitute a hierarchical visualisiton of a piece. Higher layers of the wavescape represent larger segments of music;

- Wavescapes can unveil patterns of tonal stability, tension, and harmonic qualia within a piece.

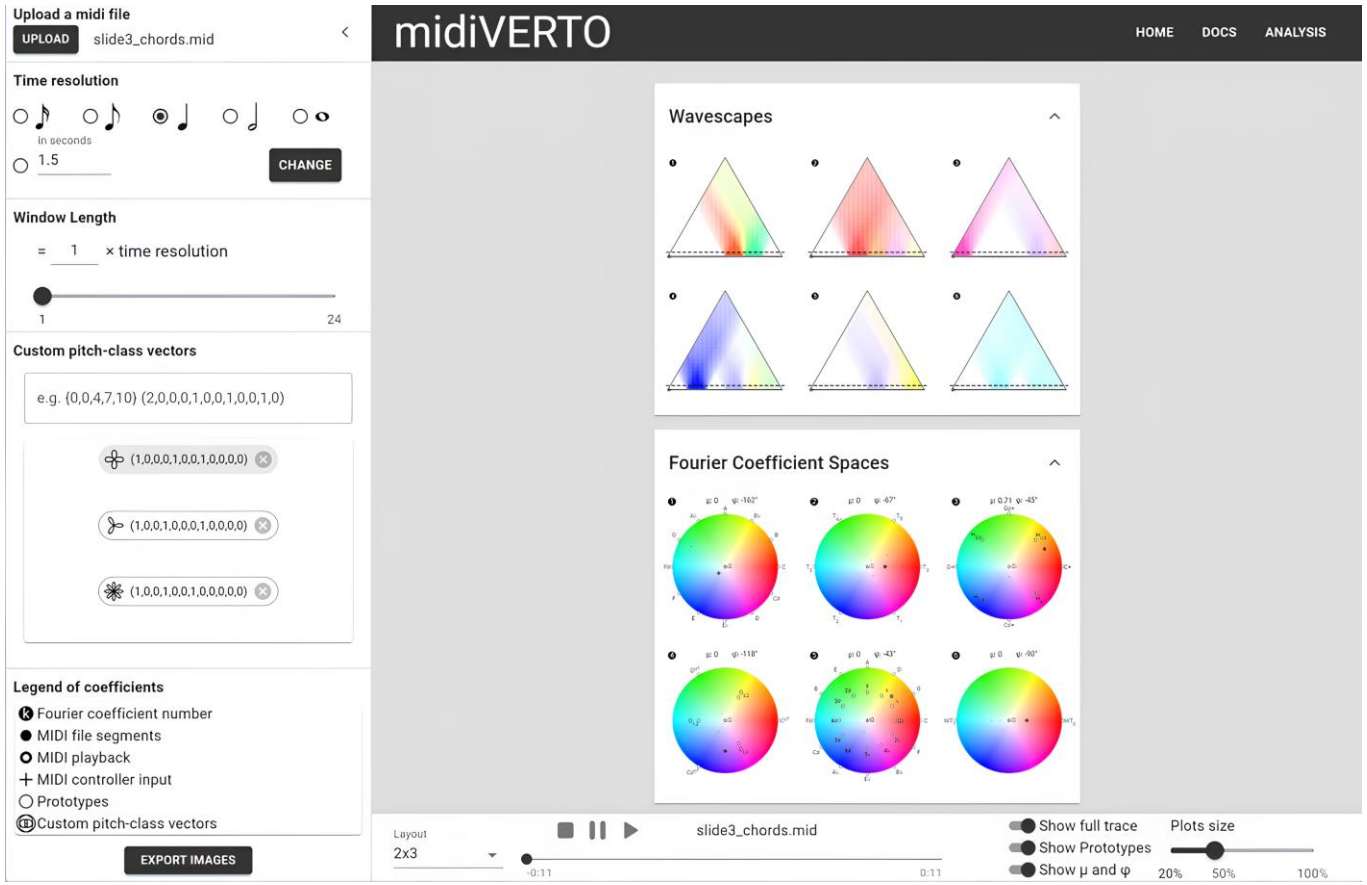


- *The triangular and hierarchical structure of a wavescape [10].*

3. The midiVERTO app.

- The midiVERTO [4] app allows users to visualize the harmonic wavescapes and phase spaces of a piece in an intuitive and accessible manner;

- The app supports interactive exploration, making it a valuable tool for analysts and researchers.



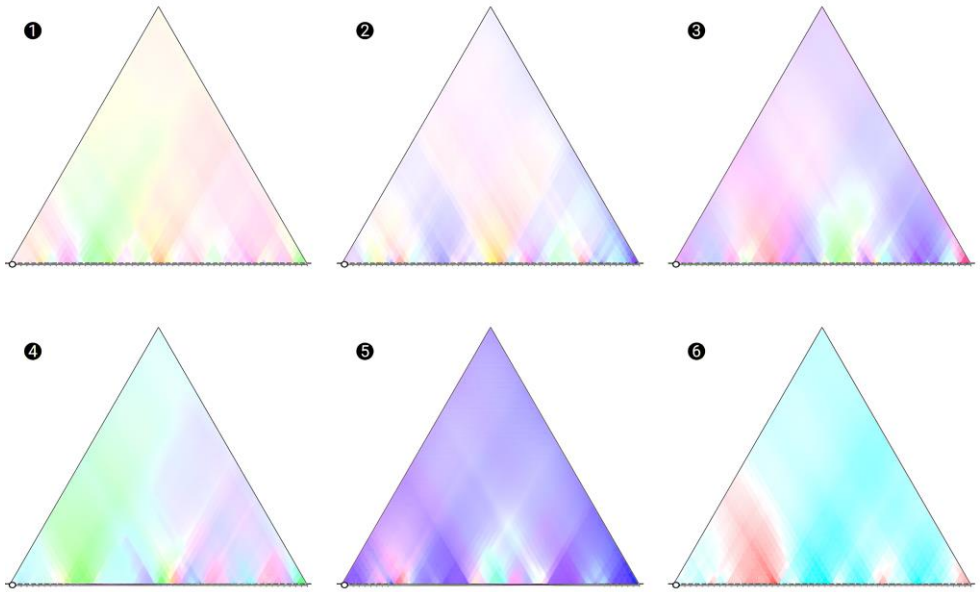
- *midiVerto's [4] interface.*

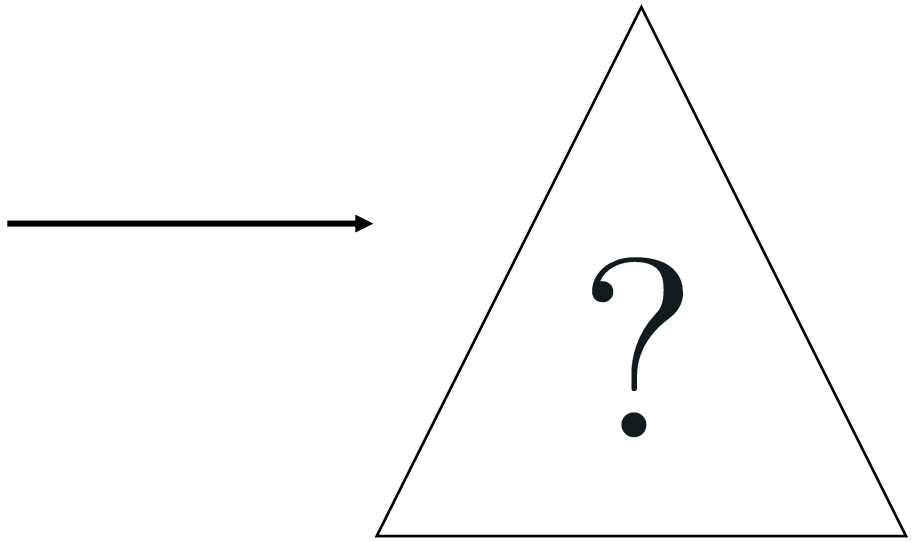
https://dcmlab.github.io/midiVERTO

4. How to improve into a single qualia wavescape?

- How to merge in one triangular structure all DFT harmonic qualities?

- How to assess with the same representation how much ambiguous a piece's qualia is?





- *The six non-trivial DFT coefficient wavescapes. [10]*

...the principle by which something has two or more mutually exclusive ways of being perceived...[9]

...interlude on qualia ambiguity...

...how much a set class is associated with a harmonic quality to the detriment of others...[6]

5. The Fourier Qualia Space (FQS).

- In the FQS [6], given the DFT magnitudes of a set class K:

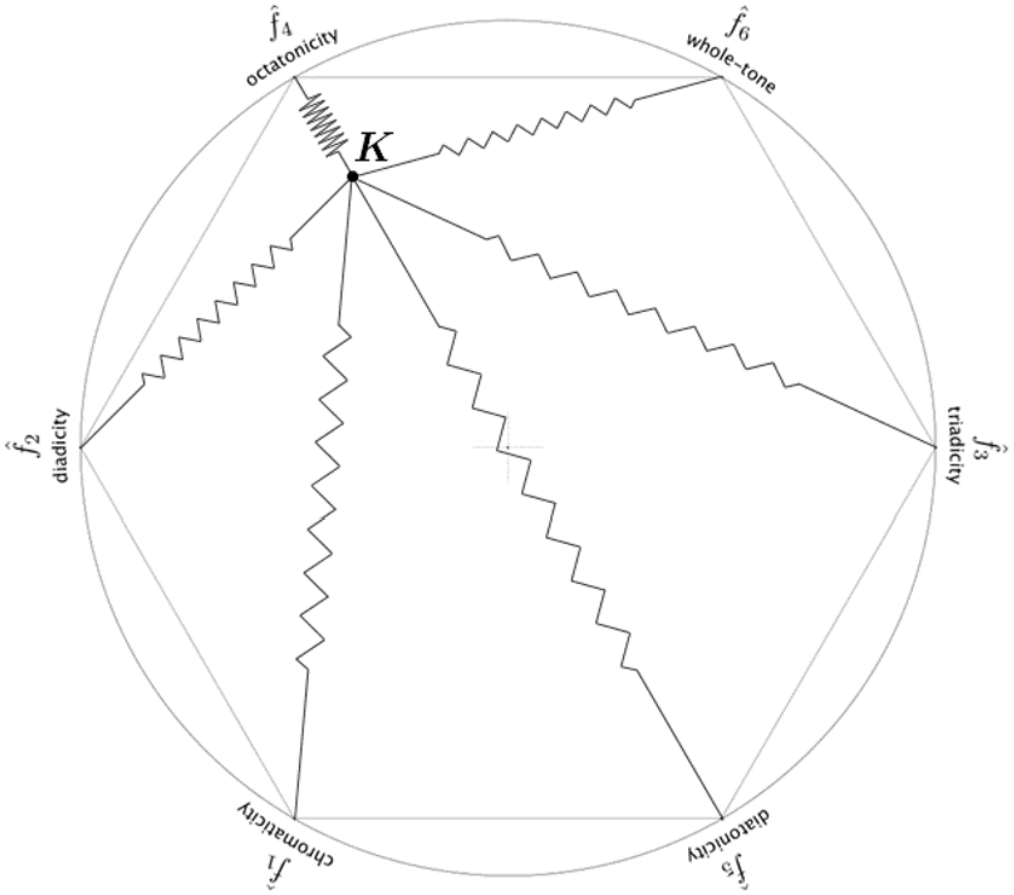
$$K = \left(\left|f_1\right|, \left|f_2\right|, \dots, \left|f_6\right|\right) \quad ,$$

its 2-dimensional coordinates can be obtained with the formula:

$$P_K = \ x, y$$

$$P_K = \left(\frac{\sum_{j=1}^{n} x_j \cdot \left|f_j\right|}{\sum_{j=1}^{n} \left|f_j\right|}, \frac{\sum_{j=1}^{n} y_j \cdot \left|f_j\right|}{\sum_{j=1}^{n} \left|f_j\right|} \right) \quad ,$$

i.e., the point where the sum of the spring forces is null.



- *RadViz's spring model for a 6D space. [3, 6]*

5. The Fourier Qualia Space (FQS).

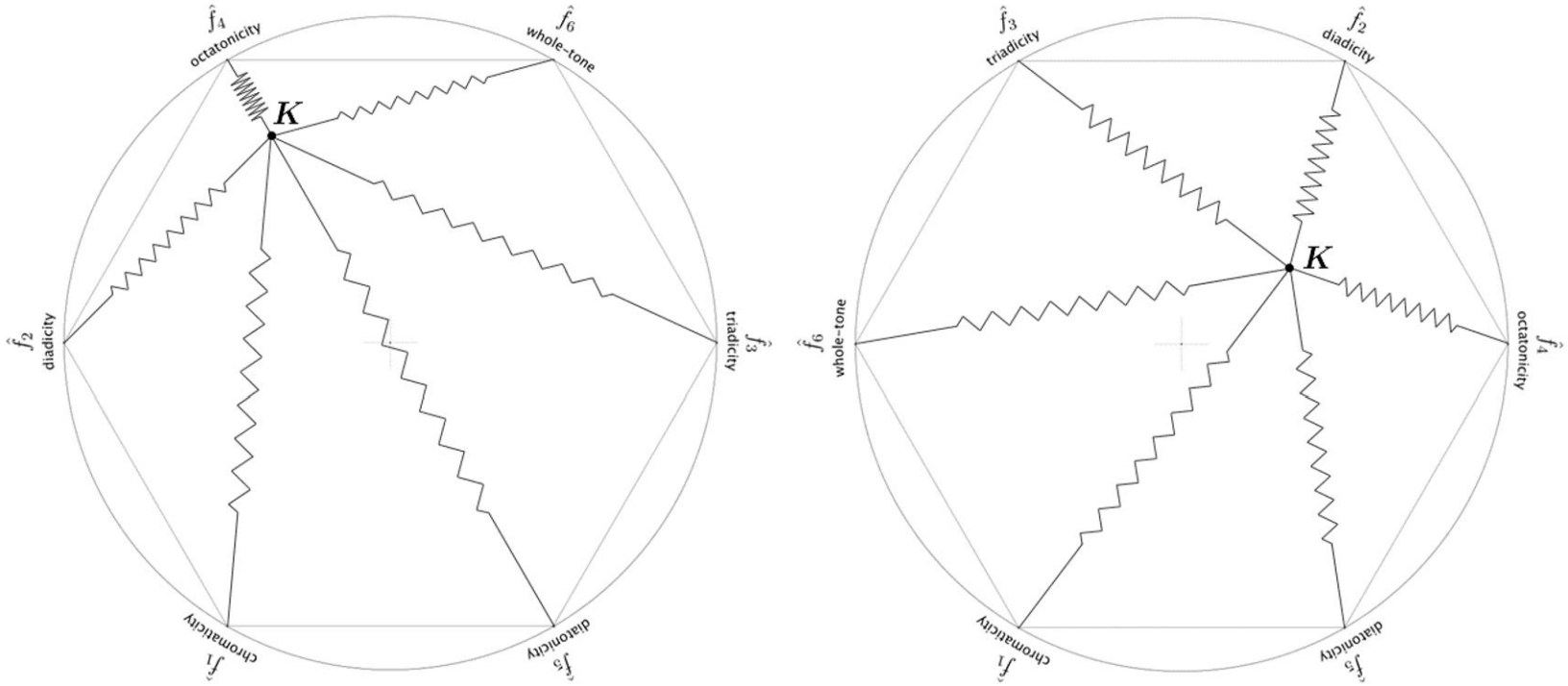
In the image, K is Stravinsky's *Petrushka* chord:

- Pitch-classes:
  $\{0,1,4,6,7,10\}$

- Set class:
  013679

- DFT magnitudes:

$$\left|f_1\right| = 0, \left|f_2\right| = 2, \left|f_3\right| = 0, \left|f_4\right| \approx 3.5, \left|f_5\right| = 0, \left|f_6\right| = 2$$



- *Two DFT coefficient orders in the RadViz circle resulting in two distinct coordinates for K. [6]*
  Left – 3, 6, 4, 2, 1, 5
  Right – 4, 2, 3, 6, 1, 5

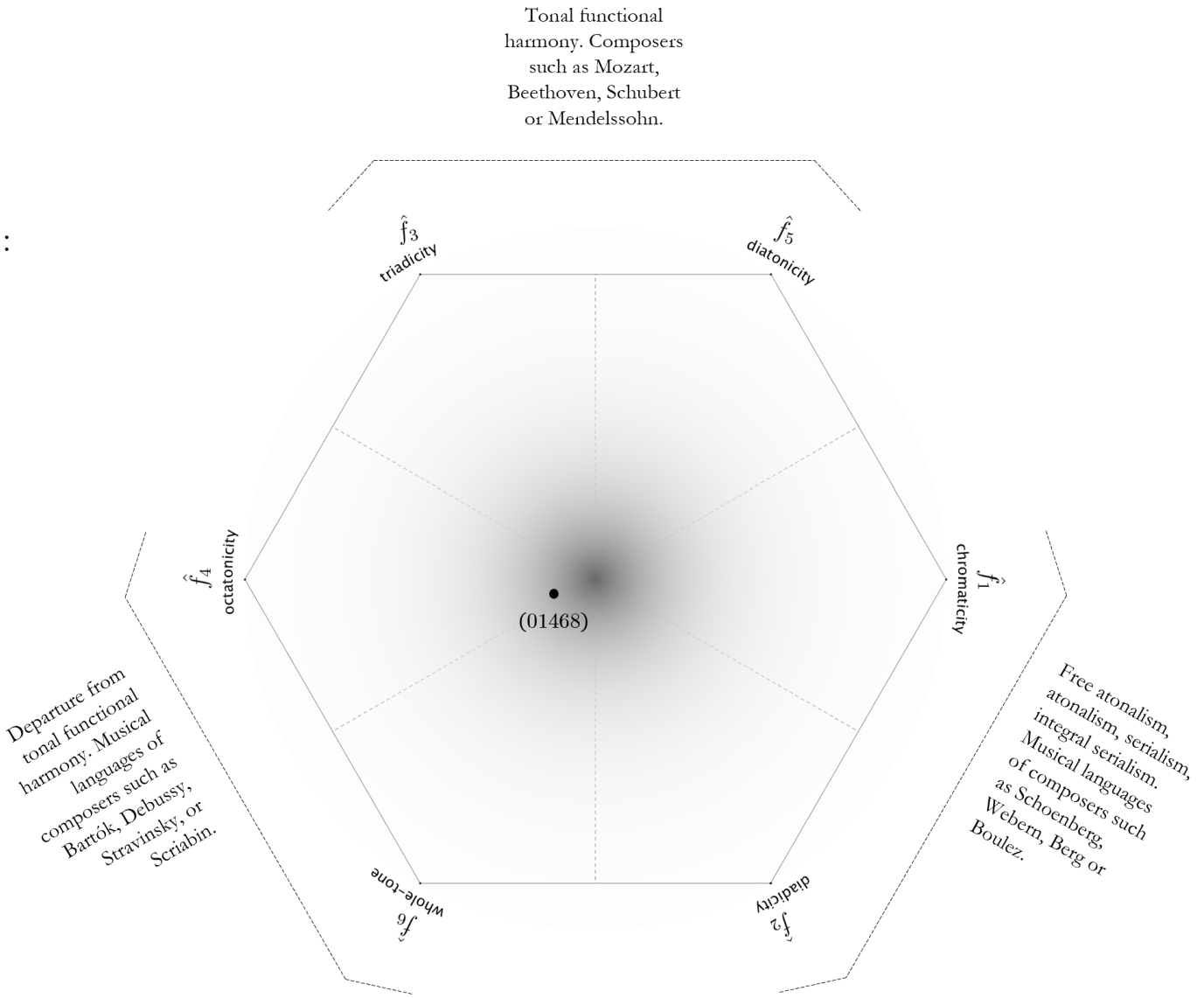
5.1. (Meta)-regions in the FQS.

- Final order [6] of DFT coefficients in the FQS:

$$f_3, f_5, f_1, f_2, f_6, f_4$$

- Inside the FQS's hexagon coexist six regions for each DFT coefficient.

- From this order arise 3 larger meta-regions linked with different musical periods and aesthetics.



- *The FQS's 6 qualia regions and the 3 meta-regions [6].*

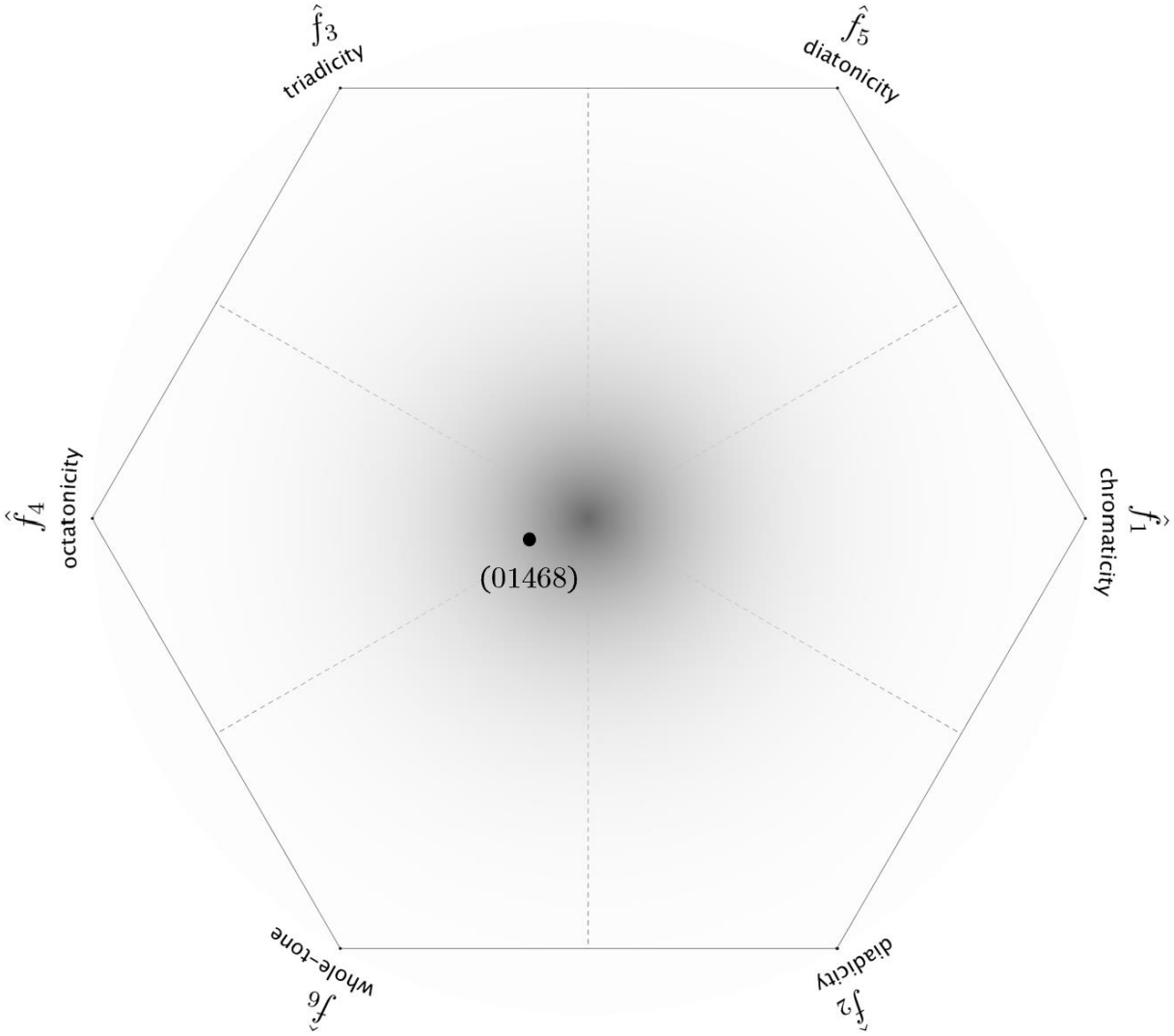
5.2. Qualia Ambiguity.

- The FQS's centre and its vicinity represent a zone of qualia ambiguity, where two or more harmonic qualities coexist with approximate intensity. [6]

- DFT magnitude profile of (01468):

$$\left|f_1\right| = 0.52, \left|f_2\right| = 1.73, \left|f_3\right| = 2.23$$

$$\left|f_4\right| = 1, \left|f_5\right| = 1.93, \left|f_6\right| = 3$$



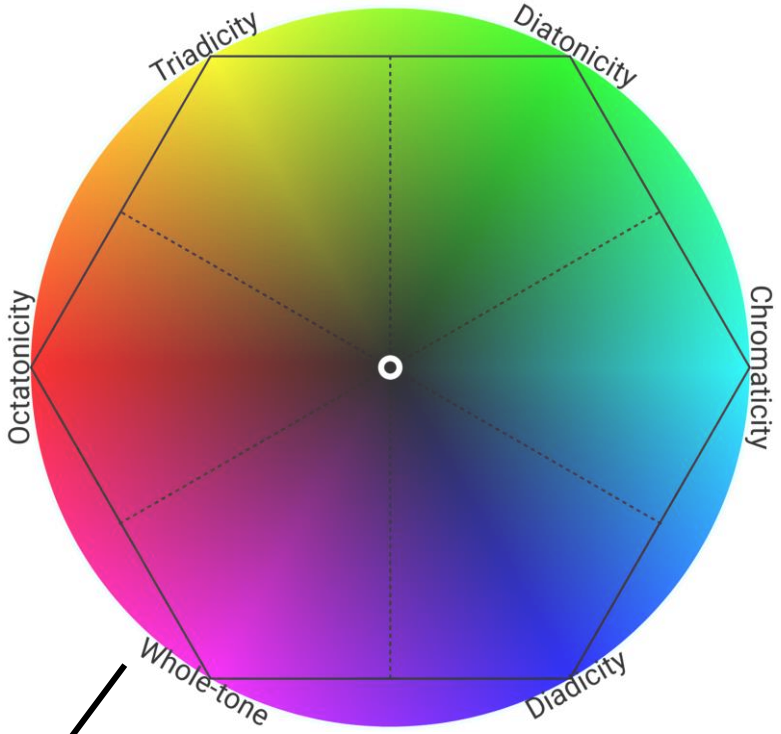
- *Qualia ambiguity in the FQS with the example of set class (01468).*

5.3. Mapping the FQS into a coloured wavescape.

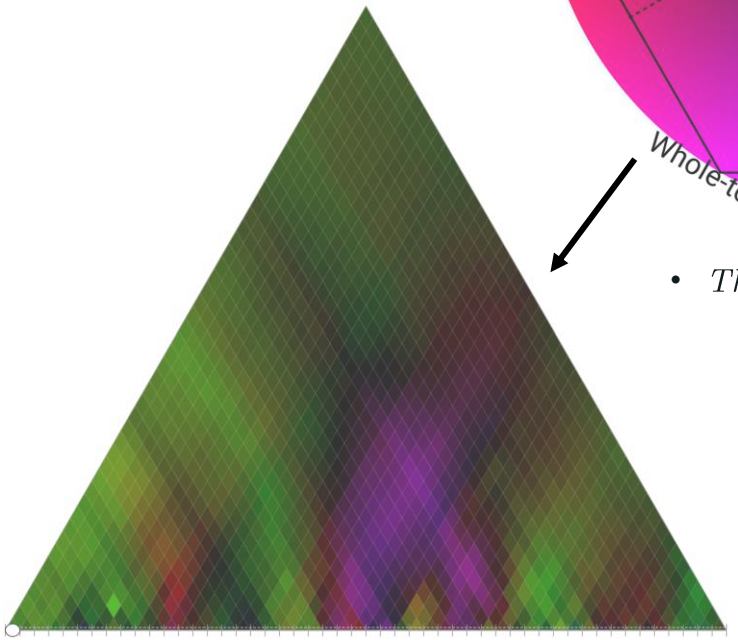
- Hue function that maps to (R, G, B).

$$h\colon [0,2\pi[\mapsto [0,1]^3$$

$$h\ \theta\ = \begin{cases} \left(0,1,1-\dfrac{3\theta}{\pi}\right), & \text{if} \quad 0 \le \theta < \dfrac{\pi}{3} \\ \left(\dfrac{3\theta}{\pi}-1,1,0\right), & \text{if} \quad \dfrac{\pi}{3} \le \theta < \dfrac{2\pi}{3} \\ \left(1,2-\dfrac{3\theta}{\pi},0\right), & \text{if} \quad \dfrac{2\pi}{3} \le \theta < \pi \\ \left(1,0,\dfrac{3\theta}{\pi}-3\right), & \text{if} \quad \pi \le \theta < \dfrac{4\pi}{3} \\ \left(4-\dfrac{3\theta}{\pi},0,1\right), & \text{if} \quad \dfrac{4\pi}{3} \le \theta < \dfrac{5\pi}{3} \\ \left(0,\dfrac{3\theta}{\pi}-5,1\right), & \text{if} \quad \dfrac{5\pi}{3} \le \theta < 2\pi \end{cases}$$



- *The FQS with hue function assigned.*



- *An example of a wavescape resulting from our method.*

...Debussy's *Reflets dans l'eau, Images*, Book I, No. 1...

6. Analytical examples.

...Bach's *Three-part Invention* (Symphony No. 9) in F minor, BWV 795...

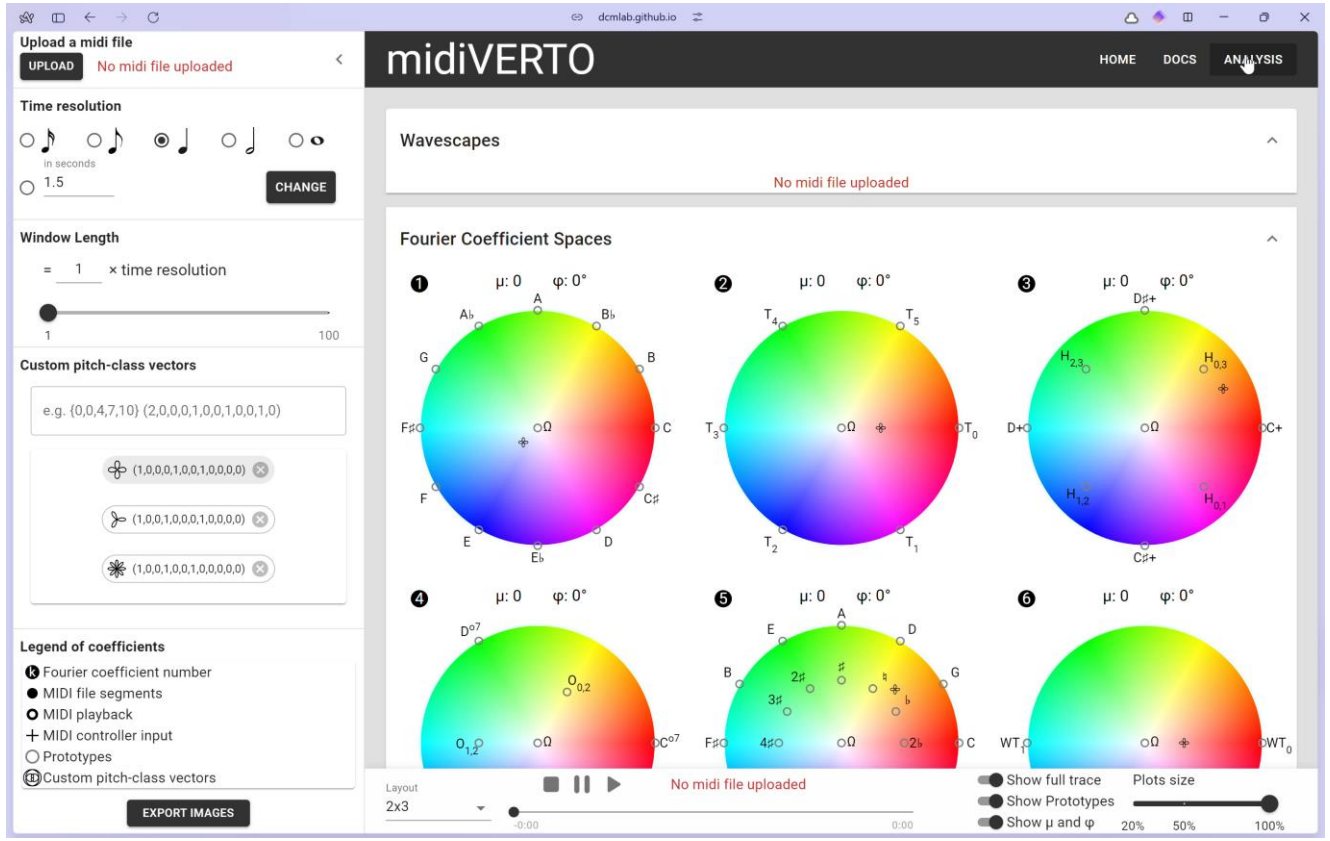
...Webern's *Four Pieces for Violin and Piano*, Op. 7, No. 1...

6.1. Bach's Three-part Invention BWV 795.

- ascending leap of a third followed by a descending second;
- motifs undergo inversion, expansion, imitation, and combination.



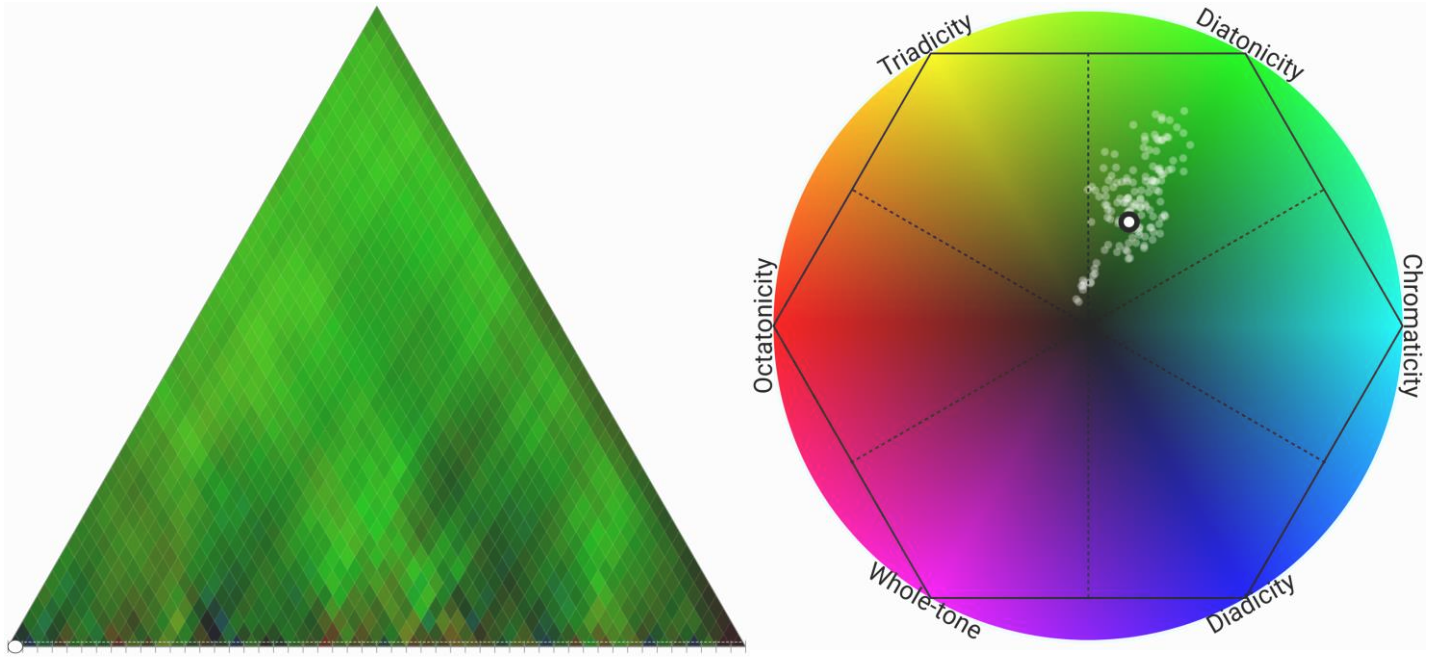
- *Opening bars of the BWV 795.*



- *Demonstration of the qualia wavescape for the BWV 795 in midiVERTO.*

6.1. Bach's Three-part Invention BWV 795.

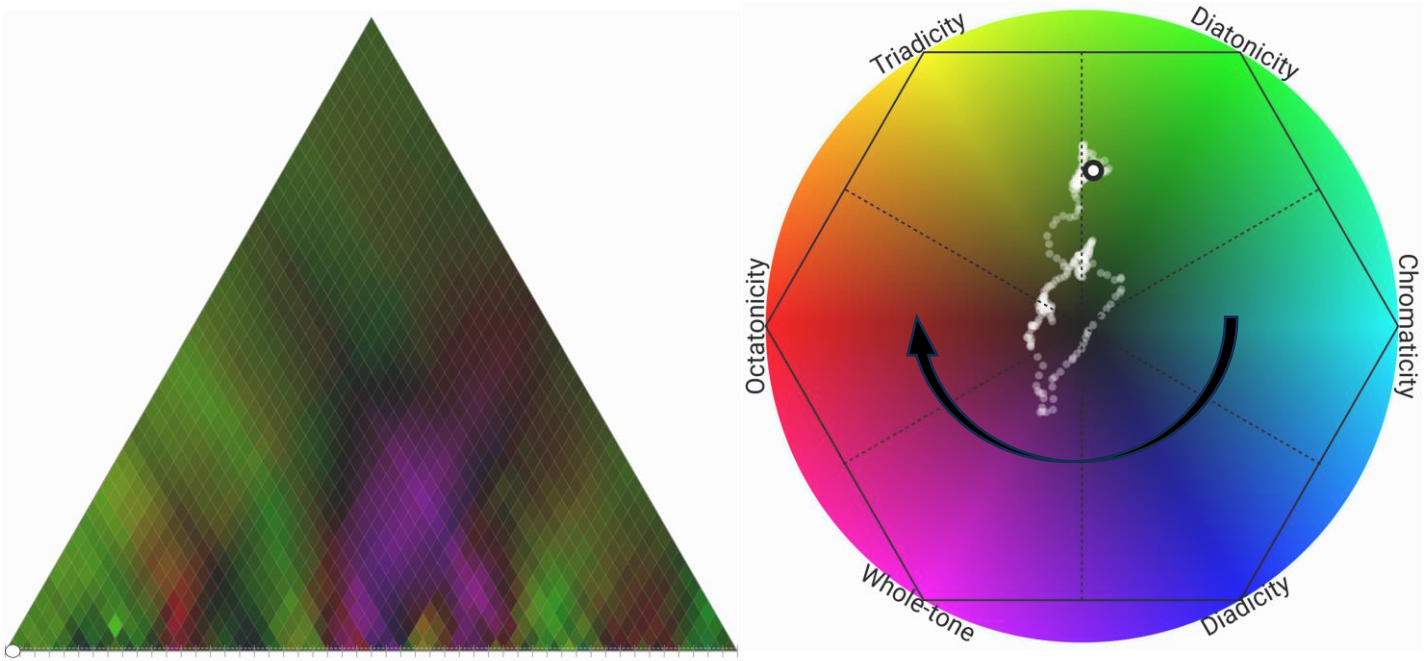
- Predominant diatonic structures;
- Low variability in the work's qualia;
- Focal points of qualia ambiguity;
- In the FQS, the piece's segments are positioned in the meta-region of functional harmony (f5 + f3);



- *Bach's Three-Part Invention wavescape, in the left, and at the right, the dispersion of the piece's segments in the FQS.*
  Time resolution *– quarter note*
  Window length *= 26*

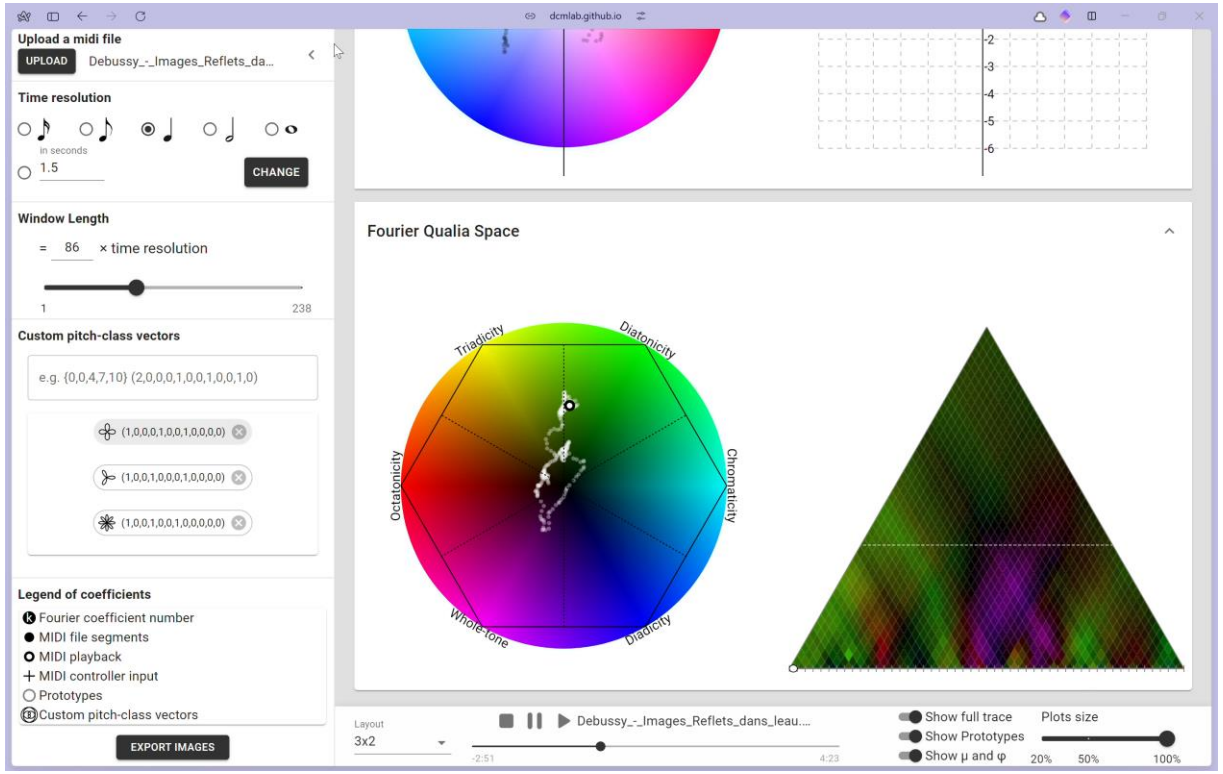
6.2. Debussy's *Reflets dans l'eau*, *Images* I/1.

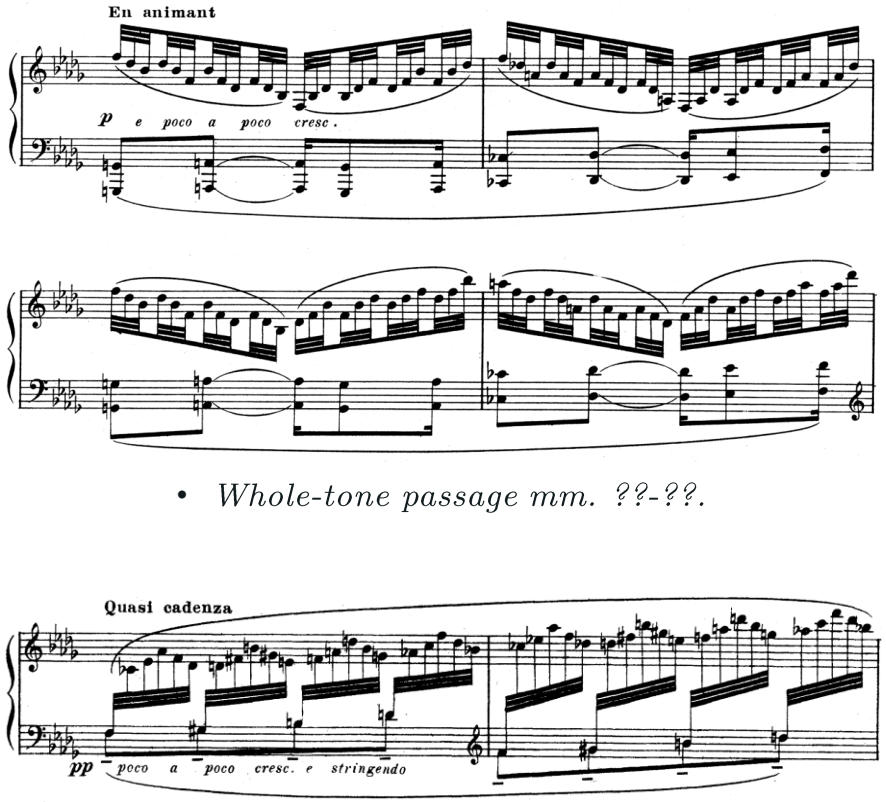
- Increased qualia complexity;
- Darker patches in higher hierarchical levels;
- Pendulum-like movement in the FQS between diatonicity and whole-tone;



- *Debussy's Reflets dans l'eau wavescape, in the left, and at the right, the pendulum movement in the FQS.*
  Time resolution *– quarter note*
  Window length *= 86*

6.2. Debussy's *Reflets dans l'eau*, *Images* I/1.

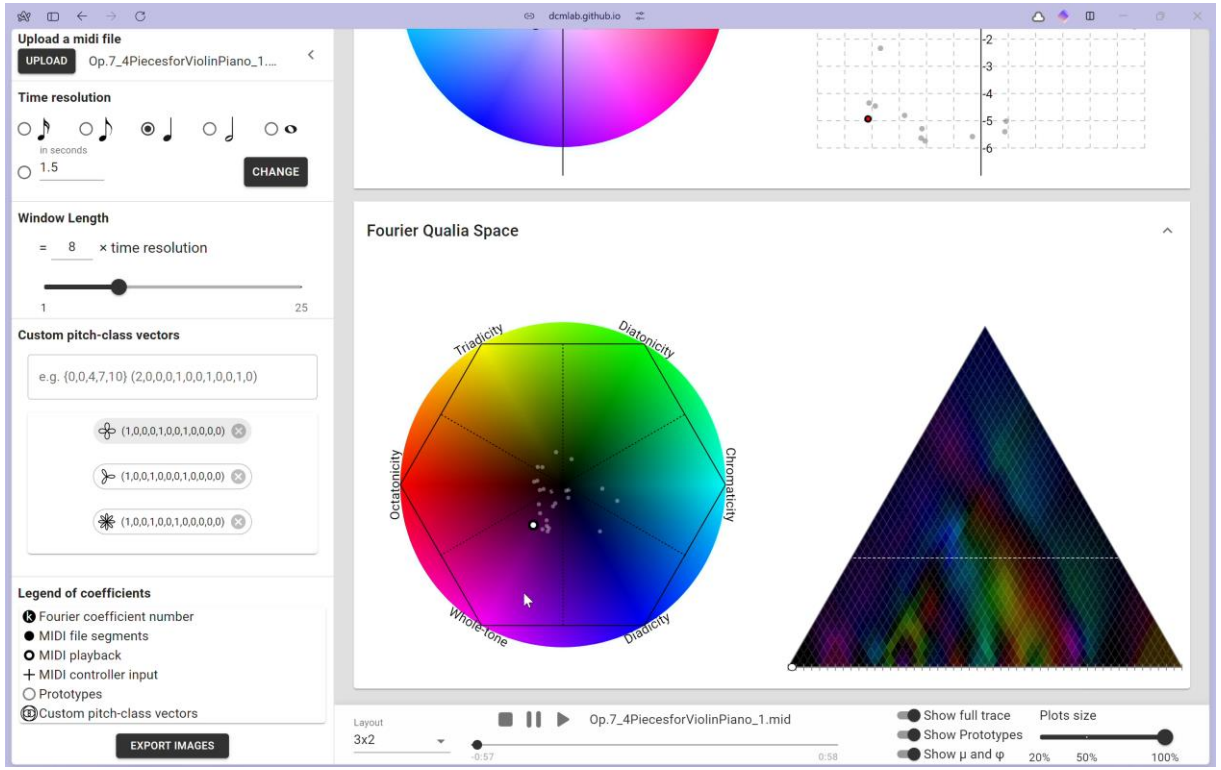


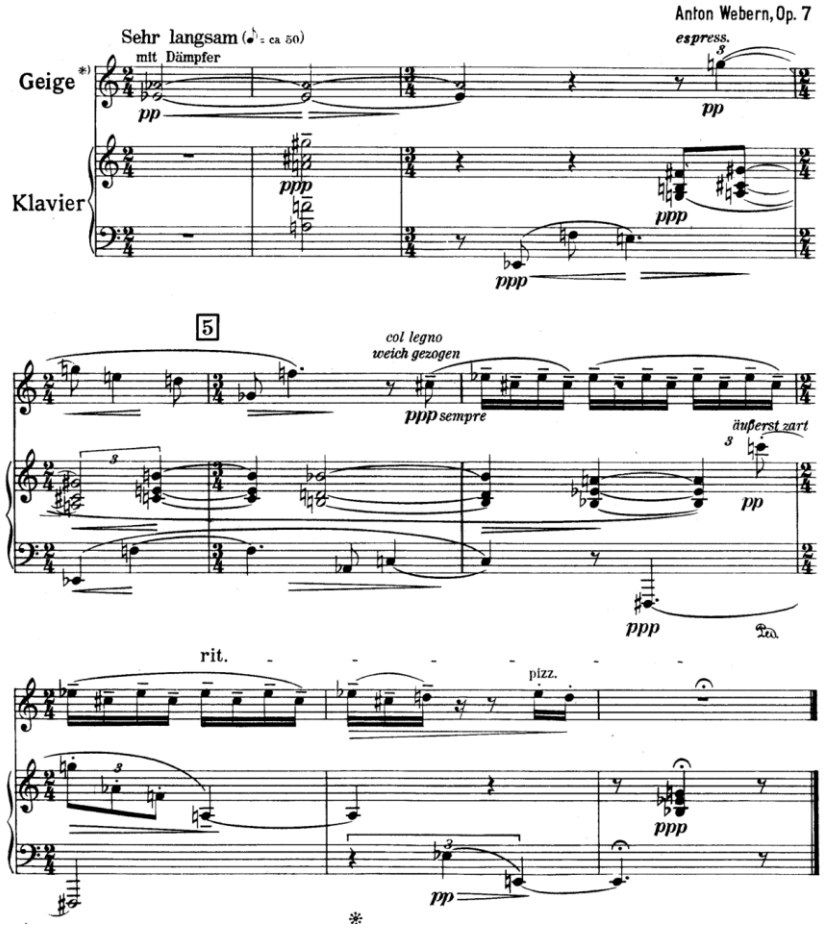


- *Whole-tone passage mm. ??-??.*

- *The seventh and ninth chords sequence responsible for the wavescape's redish section.*

6.3. Webern's Op. 7, No. 1.





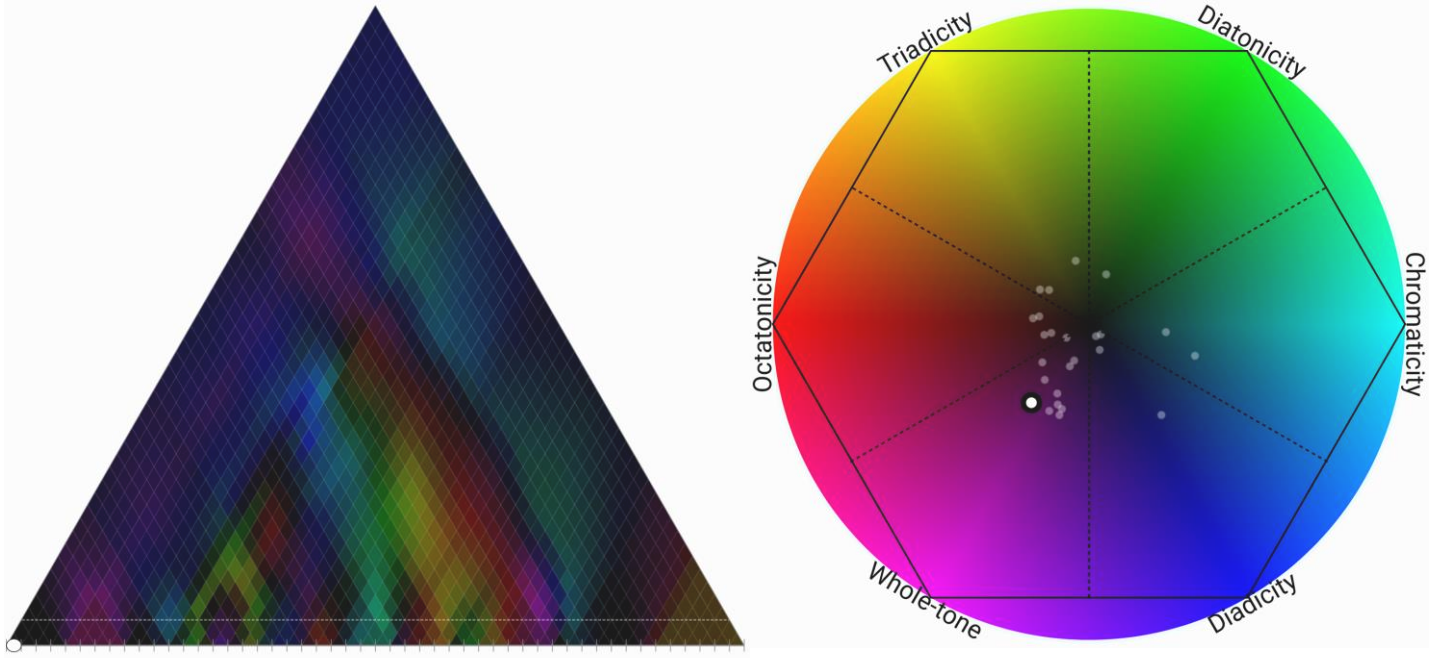
- *Webern's score of his Op. 7, No. 1, Three Pieces for Violin and Piano.*

6.3. Webern's Op. 7, No. 1.

- More generalised qualia ambiguity;

- Frequent shifts in harmonic qualities and higher dispersion in the FQS to multiple regions;

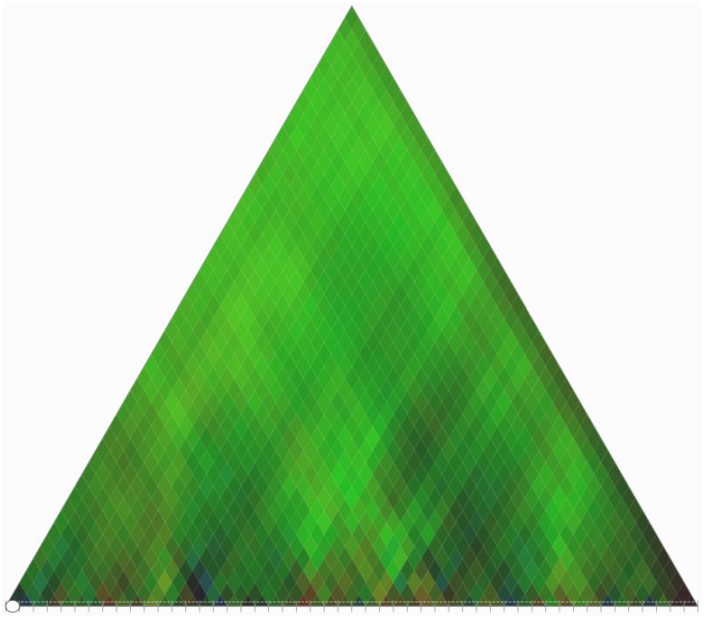
- Higher structures in the wavescape outline bluish tones linked with the atonal meta-region.



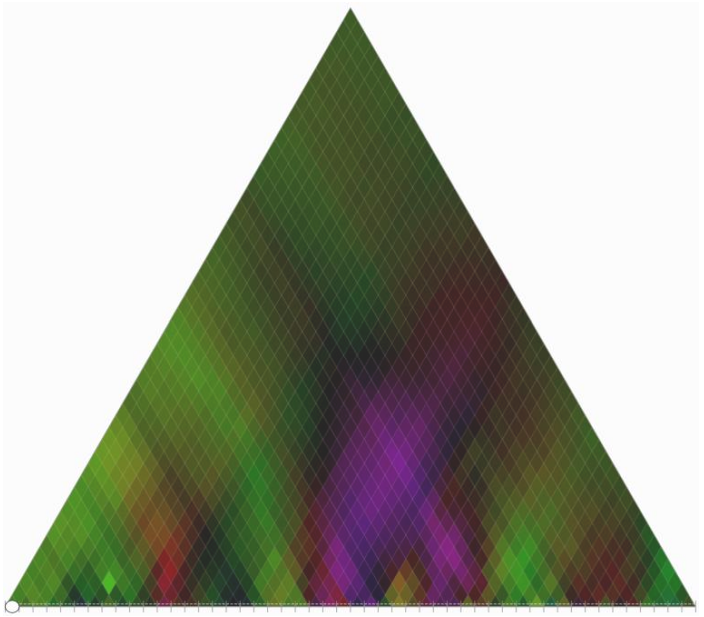
- *Webern's Op. 7, No. 1 wavescape, in the left, and at the right, the dispersion of the piece's segments in the FQS.*
  Time resolution *– quarter note*
  Window length *= 8*

7. Wavescape comparison.

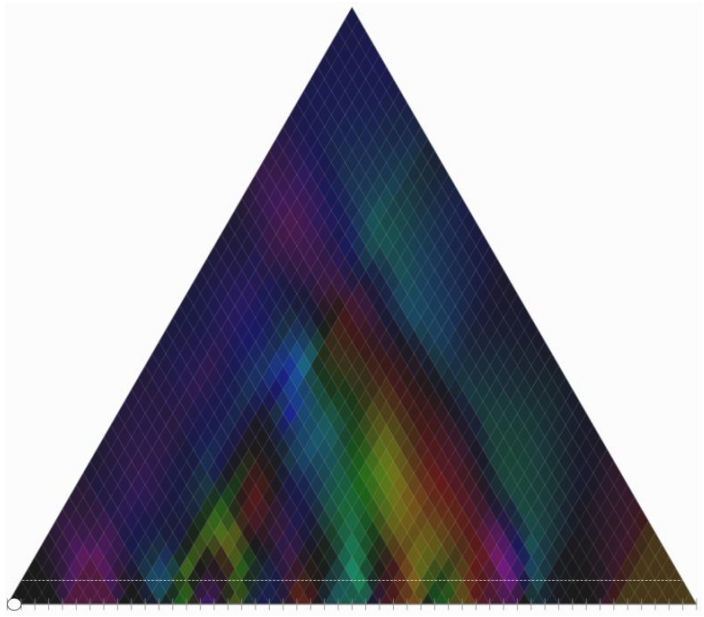
- Increased qualia ambiguity;
- Increased qualia complexity;
- Increased qualia plasticity.



- *Bach's Three-Part Invention wavescape.*



- *Debussy's Reflets dans l'eau wavescape.*



- *Webern's Op. 7, No. 1 wavescape.*

- 1723
- 1905
- 1910-1914

…a single visualisation for DFT harmonic qualities and ambiguity…

8. Conclusion.

…a versatile approach…

…available on https://dcmlab.github.io/midiVERTO…

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Thank you! ☺ **(this is a real smile!)**